General Advice

It is very important to take the HW seriously, to try the problems yourself and see how much you can do. I think this is likely a good deal more important than it might have been in lower level math classes.

In a class like Math 52, if you took that here, you likely had already taken a class somewhat like that in high school. And even if not, the nature of the content probably wasn't foreign to you. E.g. you do more advanced integrals in that class, but integrals are not new to you. In this class, the content is more likely to be completely new to many of you.

The HW, more than anything, is what will help you develop the skills/understanding to do well on the quizzes/exams. Ok that doesn't necessarily mean doing the HW guarantees success on the quizzes/exams, but it's certainly the first step towards that.

That's why it's very important you try the problems yourself, don't just copy off someone or use internet tools to do the questions for you! That might help you get the assignment done faster, but it won't help you develop the skills you need, and then you are more likely to struggle on quizzes/exams.

That incidentally is one big reason I have the quizzes, they are perhaps the best gauge of how well you are keeping up with the class. I can tell you from experience it is quite rare to have a student who is doing poorly on the quizzes suddenly do much better on the exams. So if you are struggling with the quizzes, that's definitely something to address sooner rather than later.

While doing the HW seriously is important, don't take this to mean that you shouldn't work together or come to me for help - you should. It's just, as I've alluded, I think there is kind of right and wrong way to do that.

Both in lower and higher level classes I often get people coming to office hours to ask for help with a problem. And when they do, I often try to gauge "What did you try, where did you get stuck?"

But too often their answer is "I didn't really try anything I had no idea how to start/do it." Well if you don't have any idea, look over the notes, read the book, check examples of things in the same arena.

Point is you can't give up so easily! Have to really give it an honest try. Even if in the end you don't manage to solve the problem in full and still need to ask for help, people often underestimate how much they learn from the process. Key is you should not look at it as a completely binary thing, i.e. that you either solved it or you didn't, and that if you failed to solve it completely that you got nothing out of the process - that's rarely the case! You can learn a lot from the process.

I.e. don't think that because you didn't manage to solve it in full that you wasted your time trying, and that you should have just asked for help right away. Again the effort, even if it feels like struggle in the moment, can be very useful!

Say you had no idea how to do a problem so asked me, a classmate, or some online tool how to do it, and you just outright got the answer that way. Being honest, ask yourself if that will give you the understanding that you might need to solve a problem on a quiz/exam which tests you on similar concepts.

If you are honest with yourself, I think you will generally realize the answer is no.

That is why trying the problems yourself, even if you don't manage to succeed, is so relevant! Quite unlikely that most will manage to solve every problem on their own so that's not to discourage you from asking for help,

just try to do that in the right way.

General Nature of Content

Nature of the content in this class (at least the linear algebra portion) will likely be stuff you haven't seen before, very different than calculus courses.

Many of you the most recent math course you took was probably Math 52 or 53 (Calc II, III or something roughly equivalent to one of those). In those classes, often the first thing you think in a question is "I need to take such and such derivative or such and such integral". That definitely won't be the case in the linear algebra portion of this class. You will barely, if at all, hear the terms derivative or integral for the first 70 - 80% of the class.

The differential equations part will include alot of stuff from Math 52 in contrast.

But with the differential equations portion, there will actually be some linear algebra stuff mixed in so there is a correlation certainly, part of the reason that that content is included in this class (shall allude to an example later today).

There is quite a lot of content in this class. So I won't be able to do as many examples as I like due to the theory I need to cover - examples will be more for discussion.

But also what content to present? There are many important results/theorems, should I prove them all?

Well if I do, I'll be doing nothing but proofs all of class! And that's not a good thing. So have to pick and choose a bit.

This is kind of a transition class, getting you ready to take upper division classes where proving things

becomes much more pertinent. So proving some things (and you all occasionally doing proofs yourself) is a relevant thing so that you prepare yourself a bit for higher level classes.

What will I prove? Somewhat based on the importance of the result, but bigger factor is I try to prove things where the ideas in the proof are valuable, ideas that you might use in practical problems you'd get (since you won't be doing many proofs yourself).

Proofs which use very offbeat ideas I'd likely avoid.

Be cautious about the difficulty of the material. Most classes get more difficult as the class progresses, nothing unusual in that by itself. Ok, there are perhaps exceptions, e.g. Math 52 starts off quite difficult, tough integrals, that content is, in some people's view, amongst the hardest in that class.

Ok, somewhat subjective, but in this class the material unequivocally gets much harder about a third of the way through (students say this, and exam results clearly indicate it as well).

Chapter 4 is when it gets tougher, right around when the content for Midterm 1 ends (as alluded, the averages on Midterm 2 are always much lower than Midterm 1 despite designing them with the same mindset).

This class is more extreme than most in how much tougher it gets, in Chapter 4 things get much more conceptual so not surprising. Chapters 4 - 7 use the computational ideas of Chapters 1 - 3, just in much tougher, more conceptual settings.

The content at the end of the class, the differential equations part of the class, is somewhat hit or miss.

Those who learned and remember it well from Math 52 may find it easier than the later parts of the linear algebra portion of the class.

Others might find it to be the toughest part of the class, in part because you start doing derivatives and integrals again extensively (and often not trivial integrals).

So suddenly after not having done those for the first 70 - 80% of the class, you have to switch your mindset back to doing those again - definitely hard for many!

Especially because you shifted your mindset away from those early in the course - then you have to go back. Can certainly throw people for a loop!

Chapter 1

1.1 Linear Systems

System of equations:

$$2x_1 + 3x_2 = 10$$
$$4x_1 + 11x_2 = 16$$

This specifically is a linear system, where each term with a variable has "degree" one, i.e. no squares, cubes, or variables multiplied together, nothing like any of those.

That is the main type of system we deal with in this class but in many different contexts and various motivations.

How to solve this? Two main ways:

(1) Add/subtract a constant multiple of one equation to the other to eliminate one variable.

(2) Solve one variable for other in one equation, and plug that into other equation, similarly eliminates one of the variables.

This is sort of a 2 x 2 system, since two equations and two variables. But what if it was more extreme, say ten equations in ten variables?

Not saying that would be practically reasonable to ask, but how would we try? Maybe above methods, just much longer? How can we make that as efficient as possible though?

And what if number of equations is not "in sync" with the number of variables? Very likely we'll then have infinitely many or no solutions. But how to assess that?

Two settings where this solving of linear systems can be useful are as follows. These come up much later in the class but both relate to things you've done before so useful to mention now to motivate a bit why the things we'll be doing are useful.

In three dimensions, i.e. $\mathbb{R}^3$, say we take a plane, two dimensional in nature, and a point not in it. If we wanted to "walk" from the point to the plane, what would be the fastest way to do that?

Draw a perpendicular from the point to the plane, the "foot" of that perpendicular would be the nearest point.

But how would we find that if we are given the equation of the plane and coordinates of the point in question?

Many possible ways, there is even a formula for it. But mostly based on ideas with vectors and cross products.

But now say we are in ten dimensions, e.g. $\mathbb{R}^{10}$, and we take a seven dimensional "piece" of it along with a point not in that piece. We can still define distance via distance formula in same way, ten coordinates versus three makes no difference in that.

That being so, will there still be a point in that seven dimensional piece "nearest" to that outside point?

Seems likely but how would we find it?

Well cross product is an idea that works in two or three dimensions, not higher so can't really use that.

Whether this question is practical to do by hand or not, key to doing it will be linear algebra ideas, solving a linear system.

This is in a sense one of the main things linear algebra is about. In such a setting, how can we find the closest point when the high dimension makes our earlier strategy ineffective?

Point is we will design a strategy that will work no matter how high the dimension is.

Now don't misunderstand that. That doesn't mean the calculation to solve such a problem in ten dimensions will be on the same level as in three dimensions - it definitely won't be! The point is the know how / strategy will be the same!

These problems of taking a seven dimensional piece of a ten dimensional space or solving a linear system of ten equations in ten variables are not generally going to be reasonable to do by hand, again the strategy of how to do those as efficiently as possible is the key.

If you are going to teach a machine how to do this as efficiently as possible where the calculation side of things will be no issue, well that's a big key to this.

That is one thing that makes linear algebra so important, using it in data encryption and data compression. Those both involve many layers of work and making each layer as efficient as possible is huge. And that's where linear algebra gets used a lot.

A common style calculus problem is a "mixing problem", say an aquarium with salt water that you drain/replenish to keep the liquid fresh, but naturally you have to be concerned about the concentration of salt dropping too low as the fish would then die.

So to that end we want to track how much salt is in the tank at any given moment. How do we do that?

Usually set up a differential equation which measures how much salt is entering the tank minus how much is leaving. Then solving that we find how much salt is in the tank at any given moment.

Now say there are two tanks "feeding" each other in this same kind of way. How would we measure how much salt is in each tank at a given moment?

If $x_1(t), x_2(t)$ are the amounts of salt at time t, then $x_1'(t)$ will depend on both x_1 and x_2 as will $x_2'(t)$

So then we have a system of differential equations. But how to solve that?

Well that involves using both linear algebra ideas and ideas of differential equations. Hence one big reason why this course goes over things in each, even though they seem unrelated at first - the ideas in those overlap more than you might initially think!